The Dangers of AI May Not Be What You Think!

Behind the code lies a culture: young, male, fearless. What does that bring to AI safety?

KEVIN R. HAYLETT

AUG 28, 2025

Where Does AI Risk Start?

The discourse surrounding artificial intelligence (AI) risk typically centres on technical failures, misalignment of models, or socio-political consequences like misuse or inequity. Yet a less explored dimension looms large: **the demographic composition of the machine learning (ML) field itself**.

The practitioners, who are predominantly young men aged between 21 and 30, shape not only the technology but also the way risks are perceived, framed, and prioritized. it seems, to me, this demographic skew, coupled with institutional amplifiers like universities and startups, creates a structural blind spot. This blind spot may pose a greater threat than the algorithms themselves. By examining the age and gender distribution of ML researchers, the neurobiological underpinnings of risk perception, and the cultural narratives inflating AI's perceived maturity, this article proposes a new axis of AI risk analysis, namely: **Demographic Risk**.

A Youth-Dominated Field:

The ML Cohort

My own observations and research from across the internet, is far from complete and limited by the available data. However, its strongly suggests that the ML field is characterized by a striking demographic profile: a youth bulge of practitioners, mostly male, with a median age likely between 25 and 30. My rough and ready preliminary research, suggests that approximately 80% of AI professionals have less than 10 years of experience, implying that the majority are in their 20s to early 30s. Gender representation appears to be equally skewed, with women comprising only about 22% of the global AI workforce, 12% of ML-specific roles, and

10–18% of authors at top AI conferences like NeurIPS or ICML. This homogeneity i.e. young, male, and early-career profiles, defines the creative space through which AI is developed and its risks assessed.

A pivotal moment in AI's recent history, the 2017 publication of "Attention Is All You Need," illustrates this dynamic. The paper, which introduced the transformative Transformer architecture, was co-authored by a team including Aidan Gomez, then a 20-year-old Google intern, and Ashish Vaswani, approximately 31 at the time. While precise ages for all authors are unavailable, the profile of early-career researchers and interns suggests a team weighted toward youth. This snapshot reflects a broader trend: that is at the time of writing the ML field is driven by individuals under 35, many fresh from university programs that scaled rapidly after the deep learning boom of 2012. These programs, fuelled by industry demand and academic opportunism, produce thousands of graduates annually, most of whom enter the field with less than a decade of professional experience.

This demographic composition is not merely descriptive as it shapes the field's priorities. Young practitioners, steeped in a culture of rapid prototyping, conference deadlines, and venture capital cycles, focus on novelty, performance, and immediate impact. Their risk horizon may be short, cantered on optimization challenges or benchmark gains, rather than systemic or intergenerational consequences. In contrast, older researchers (40s–60s), with lived experience of technological cycles such as the AI winters of the 1980s, or the biotech's regulatory battles. This experience may offer a longer risk horizon, informed by caution and historical memory. The absence of this perspective in a youth-dominated field may be creating a structural gap in how AI risks are conceptualized and articulated.

The Neurobiology of Risk: Youth and Fearlessness

Importantly, the youth bulge in ML is not just a matter of inexperience; it is a neurobiological phenomenon. Risk perception evolves with age, mediated by brain development and life stage. The prefrontal cortex, responsible for impulse control, long-term planning, and risk assessment, does not fully mature until the mid-20s, later in men than women. In young men, aged 18–25, testosterone peaks amplify impulsivity and sensation-seeking while dampening fear responses. This is why militaries historically recruit young men as their neurological profile makes them fearless, and more willing to charge into danger that older individuals instinctively avoid.

In the context of ML, this translates to a cohort that prioritizes innovation over stability. For a 21-year-old coding a new model, “risk” is a maybe a hollow signifier, a term used in papers but not yet imbued with the weight of lived consequences; such as family responsibilities, professional failures, or societal impacts. As one observer put it, risk for this cohort is a “fractal, word path” in language, and maybe not a deeply felt reality. Older researchers, by contrast, have carved deeper semantic valleys for “risk,” shaped by experiences of technological hype cycles, regulatory friction, or personal stakes like raising children.

> **This generational divide is stark**: a field dominated by individuals under 30 may lack the neurological and experiential scaffolding to fully grasp long-term or systemic dangers.

However, this fearlessness of youth, while ideal for rapid innovation, may become a liability when building systems with global implications. The ML cohort’s primary focus is on technical puzzles, aligning objectives, reducing bias, improving interpretability. This recasts recasts risk as an engineering problem, side-lining broader concerns like societal inequity, ecological impact, or existential threats. The possible absence of a mature risk perception may be compounded by the field’s gender skew. For example, the underrepresentation of women may also mean underrepresentation of perspectives that, for a variety of sociological and experiential reasons, often prioritize caution, user-centric design, and ethical implications in tech discourse.

The Illusion of Maturity: Institutional Amplifiers

The ML field’s demographic skew may also be being amplified by institutional structures that create a false sense of maturity. Since the deep learning renaissance of 2012, many universities have rushed to establish AI/ML programs, packaging them as gateways to an “AI revolution.” These new programs produce graduates who may gte an impression they are entering a field as robust and cumulative as medicine or civil engineering. Yet, as many have notes, the foundations of AI remain fragile and contested. Leading researcher debate even the basics of how LLMs behaviours emerge. Additionally, there is a torrent of research papers. Thousands of ML papers are published annually at conferences like NeurIPS. It seems to me that this is creating an illusion of a solid well established field, that is evidentially not true. However, much of this output, and research, is incremental or speculative, chasing benchmarks rather than addressing foundational gaps and understanding.

This torrent of papers may be mistaken for maturity, particularly by a cohort lacking historical context. Unlike those scientists who witnessed the AI winters of the 1980s or the slow regulatory grind of biotech, young ML practitioners have only experienced the post-2012 hype cycle. It's quite possible that they may conflate volume with depth, assuming the field's rapid output signals a stable, well-understood discipline.

Meanwhile, real-world indicators belie this perception: medical devices still take a decade to reach market, life expectancy has plateaued, and job creation versus replacement remains a political tug-of-war. It seems to me that the human condition has not quite shifted at the exponential rate of conference submissions.

Universities and industry exacerbate this illusion.

It's maybe unsurprising academic programs market AI as inevitable: more students mean growth and more resources. At the same time start-ups and venture capital reward speed and novelty over caution.

> These factors maybe creating a feedback loop: a youthful, risk-blind cohort is incentivized to produce more, faster, without questioning the field's underlying stability or addressing critical real world safety issues.

The result appears to be a narrative of exponential progress that may be masking the slow, messy reality of deploying technology in human systems.

Sociological Risk: A Field That Can't See Itself

Following these lines of thought, we can then see that the demographic homogeneity of ML practitioners does not merely limit risk perception: it distorts the entire AI risk discourse. The following three dynamics appear to stand out:

Optimism Bias:

Firstly, young men, driven by testosterone and early-career ambition, lean toward overconfidence. They frame risks as solvable technical challenges, not intractable societal problems. This aligns uncomfortably with the hype cycles of AI, where breakthroughs are celebrated as steps toward a new utopia rather than scrutinized for unintended consequences and real risks to society.

Temporal Myopia:

Secondly, with less than a decade of experience, most practitioners struggle to conceptualize 20–50-year risk horizons, the bigger picture. The existential risks, like runaway AGI, capture attention maybe because they are speculative and exciting, fitting the cohort's novelty-seeking mindset. Meanwhile, slow-burn risks, such as healthcare bottlenecks, economic erosion, or sociotechnical instability, receive less focus, as they require a longer temporal perspective the cohort lacks.

Echo Chamber:

Thirdly, a field dominated by young men from elite, often Western institutions may reinforce its immediate values. Alternative perspectives, such as feminist ethics, ecological considerations, or historical analogies, are easily side-lined. Disciplines like nonlinear dynamical systems, which could offer critical insights into complex system behaviours, remain "invisible" to the ML mainstream, further narrowing the risk lens.

Here we can see that the military analogy is particularly compelling. Just as young men are ideal soldiers due to their fearlessness, they are ideal innovators for the same reason. But when building systems with global stakes, this fearlessness may become a liability. A liability to the greater society. The real risk, the may not be some Sci-Fi inspired runaway AGI but a simple demographic myopia i.e. a global infrastructure of intelligent systems built by a cohort that may be structurally inclined to underweight long-term dangers.

Is This Being Considered?

Researching these ideas, I found that, remarkably, this demographic dimension of AI risk is almost entirely absent from current governance discussions. The focus of AI Risk and Safety remains consistently on technical alignment (making models safe and interpretable), socio-political risks (misuse or inequity), or diversity in terms of gender equity and dataset fairness. The cognitive and life-stage limitations of the practitioner base are rarely, if ever, addressed. And even even when diversity is discussed, it centres on representation rather than the deeper issue of cognitive diversity across age groups. The implicit assumption is that if the code works, the risk discourse is neutral. Yet, as the analysis presented here suggests, the software is being written by a cohort whose neurological and sociological profile may be skewing their perception of danger.

A Measurable Risk

Interestingly, this oversight is measurable. Research on risk perception in other domains such as adolescent car accidents, substance use, or military behaviour all show that young men, aged 18–25, exhibit peak risk-blindness. And this inability to see declines around the age of 30 as responsibilities and experience accumulate. The ML field, perched in this age bracket, it would seem likely mirrors this pattern. Yet AI governance frameworks rarely account for the practitioner's cognitive stage and assumes that technical rigour alone can address systemic and long term risks.

A New Axis: Demographic Risk

Following from the prior discussion we can build a wider framework for AI risk and add a third axis to the conventional categories:

- **Technical Risk**: Model failures, misalignments, or unintended behaviours.
- **Socio-political Risk**: Misuse, authoritarian applications, or economic dislocation.
- **Demographic Risk**: The age, gender, and life-stage biases of practitioners, which shape what risks are seen and prioritized.

If we now consider this tripartite of risks we can se it may explains why AI risk debates often oscillate between apocalyptic scenarios (AGI doomsday) and naïve optimism (faith in technical fixes). Both extremes reflect the youthful, male cohort's worldview: speculative, optimistic, with short-horizons. By contrast, older researchers or those from underrepresented groups might emphasize deeper, more systemic risks such as healthcare inequity or ecological impacts. The evaluation of such risks may require a longer temporal lens or broader inter-disciplinary perspective.

Historical Parallels: Lessons from Biotech and Nuclear Science

To contextualize this, consider historical parallels. The Manhattan Project's scientists, with an average age in their 30s–40s, focussed on the immediacy of the need for a vital weapon to end the war. Time for considering the existential stakes of nuclear technology in wartime was limited. However, after the war and development of the first atomic bombs, their caution, informed by life experience and the weight of global consequences, led to later calls for governance. Especially as they became more aware of the extreme future risks to mankind.

Similarly, biotech's Asilomar conference in 1975, driven by scientists with a broader temporal perspective, established guardrails for genetic engineering. In contrast, AI's youth-dominated cohort, trained in a high pressure output-driven environment, appears to lack this cautionary tradition. The absence of an "AI winter" in their lived experience may reinforce a belief in inevitable progress, unmoored from the regulatory or ethical frictions that tempered earlier technological revolutions.

Toward a Solution: Bridging the Demographic Gap

The questions become about, what can we do, and what should we do? In practice addressing this demographic risk may require structural interventions and these may encounter considerable friction within the industry. However, the following points are offered to frame further discussion:

- **Cross-Generational Governance -** Integrate older researchers, ethicists, and systems thinkers into AI development and policy. Their longer risk horizons can counterbalance youthful optimism.
- **Amplifying Diverse Voices -** Increase representation of women and non-Western perspectives, which may emphasize care, equity, or ecological impacts overlooked by the current cohort.
- **Institutional Reform -** Universities and conferences should prioritize long-term risk education, drawing on historical case studies like biotech or nuclear governance, rather than perpetuating the myth of AI's maturity.
- **Risk Discourse Analysis -** Study how "risk" is defined in AI literature versus older fields. A narrower, more technical framing in ML papers would confirm the cohort's limited semantic map for risk.

The Semantic Void of "Risk"

The concept of "risk" itself may also reveal the cohort's blind spot. For young ML practitioners, "risk" is a buzzword, used in papers and talks but lacking visceral depth. It seems to me its simply a "fractal, word path" not yet grounded in lived experience. Researchers over 30 with memories of technological failures or personal responsibilities, begin to imbue "risk" with multidimensional meaning including, societal, ethical, ecological and long term considerations. This **semantic void**, rooted in the cohort's youth, may underscore the need for a broader risk framework.

Onwards: AI as a Demographic Mirror

So if you've followed this far you may now see that risk of AI lies not only in its algorithms but in the demographic monoculture building them. A field dominated by young men, aged 21–30, driven by short-term incentives and fearless optimism, who may not fully perceive the long-term or systemic dangers of its creations. Institutional narratives such as university programs, conference paper gluts, and venture capital cycles all amplify this blind spot. And this creates an illusion of maturity that masks AI's fragility. By introducing **Demographic**

Risk as a new axis of analysis, alongside technical and socio-political risks, the field can begin to address this structural flaw. **Until then, AI remains a mirror of its makers: fearless, ambitious, and perilously blind to the horizons beyond their own.**

Copyright © Kevin R. Haylett 2025